

1. MANIPULATION OF NUMBERS USING READLINE STATEMENT

```
using System;

class Program
{
    static void Main()
    {
        string name;
        int m1, m2, m3, total;
        double average;

        Console.Write("Enter name: ");
        name = Console.ReadLine();

        Console.Write("Enter mark 1: ");
        m1 = Convert.ToInt32(Console.ReadLine());

        Console.Write("Enter mark 2: ");
        m2 = Convert.ToInt32(Console.ReadLine());

        Console.Write("Enter mark 3: ");
        m3 = Convert.ToInt32(Console.ReadLine());

        total = m1 + m2 + m3;
        average = total / 3.0;

        Console.WriteLine("\nName: " + name);
        Console.WriteLine("Mark 1: " + m1);
        Console.WriteLine("Mark 2: " + m2);
        Console.WriteLine("Mark 3: " + m3);
        Console.WriteLine("Total: " + total);
        Console.WriteLine("Average: " + average);
        Console.ReadLine();
    }
}
```

Output:

Enter name: Mithun

Enter mark 1: 80

Enter mark 2: 75

Enter mark 3: 90

Name: Mithun

Mark 1: 80

Mark 2: 75

Mark 3: 90

Total: 245

Average: 81.66666666666667

2. MANIPULATION OF NUMBERS USING COMMANDLINE STATEMENT

```
using System;
using System.Collections.Generic;
using System.Text;
namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            int n1 = Convert.ToInt32(args[0]);
            int n2 = Convert.ToInt32(args[1]);
            Console.WriteLine("Addition: " + (n1 + n2));
            Console.WriteLine("Subtraction: " + (n1 - n2));
            Console.WriteLine("Multiplication: " + (n1 * n2));
            Console.WriteLine("Division: " + ((double)n1 / n2));
            Console.ReadKey();
        }
    }
}
```

Output:

Addition: 15

Subtraction: 5

Multiplication: 50

Division: 2

3. BOXING AND UNBOXING

```
using System;
using System.Collections.Generic;
using System.Text;
namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            int num = 23;

            // Boxing
            object obj = num;
            Console.WriteLine("BOXING");
            Console.WriteLine("Value = " + obj);

            // Unboxing
            object obj1 = 66;
            int num2 = (int)obj1;
            Console.WriteLine("UNBOXING");
            Console.WriteLine("Value = " + num2);
            Console.ReadKey();
        }
    }
}
```

Output:

BOXING

Value = 23

UNBOXING

Value = 66

4. ODD OR EVEN NUMBERS USING IF...ELSE

```
using System;
using System.Collections.Generic;
using System.Text;
namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            Console.Write("Enter a number: ");
            int num = Convert.ToInt32(Console.ReadLine());
            if (num % 2 == 0)
            {
                Console.WriteLine("Even number");
            }
            else
            {
                Console.WriteLine("Odd number");
            }
            Console.ReadKey();
        }
    }
}
```

Output:

Enter a number: 8

Even number

Enter a number: 7

Odd number

5. GRADING SYSTEM USING SWITCH CASE

```
using System;
using System.Collections.Generic;
using System.Text;
namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            Console.Write("Enter grade: ");
            char grade = Convert.ToChar(Console.ReadLine().ToUpper());
            switch (grade)
            {
                case 'S':
                    Console.WriteLine("Excellent");
                    break;
                case 'O':
                    Console.WriteLine("Good");
                    break;
                case 'A':
                    Console.WriteLine("Fair");
                    break;
                case 'B':
                    Console.WriteLine("Average");
                    break;
                case 'M':
                    Console.WriteLine("Poor");
                    break;
```

```
        default:
            Console.WriteLine("Fail");
            break;
    }
    Console.ReadKey();
}
}
```

Output:

Enter grade: s

Excellent

6. SUM OF DIGITS USING FOR LOOP

```
using System;
using System.Collections.Generic;
using System.Text;
namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            Console.Write("Enter a number: ");
            int n = Convert.ToInt32(Console.ReadLine());
            int sum = 0;
            for (; n != 0; n = n / 10)
            {
                sum = sum + n % 10;
            }
            Console.WriteLine("Sum of digits = " + sum);
            Console.ReadKey();
        }
    }
}
```

Output:

Enter a number: 1234

Sum of digits = 10

7. ARRAYS

```
using System;
using System.Collections.Generic;
using System.Text;
namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            // 1. SINGLE DIMENSIONAL ARRAY
            Console.WriteLine("1. SINGLE DIMENSIONAL ARRAY");

            string[] a = { "abc", "lmn", "xyz" };

            for (int i = 0; i < a.Length; i++)
                Console.WriteLine(a[i]);

            // 2. SINGLE DIMENSIONAL ARRAY USING FOREACH
            Console.WriteLine("\n2. SINGLE DIMENSIONAL ARRAY - FOREACH");

            string[] b = { "pop", "up", "this" };

            foreach (string x in b)
                Console.WriteLine(x);

            // 3. TWO DIMENSIONAL ARRAY
```

```
Console.WriteLine("\n3. TWO DIMENSIONAL ARRAY");
```

```
int[,] a1 = new int[2, 2];
```

```
int[,] a2 = new int[2, 2];
```

```
Console.WriteLine("Enter Matrix A");
```

```
for (int i = 0; i < 2; i++)
```

```
    for (int j = 0; j < 2; j++)
```

```
        a1[i, j] = Convert.ToInt32(Console.ReadLine());
```

```
Console.WriteLine("Enter Matrix B");
```

```
for (int i = 0; i < 2; i++)
```

```
    for (int j = 0; j < 2; j++)
```

```
        a2[i, j] = Convert.ToInt32(Console.ReadLine());
```

```
Console.WriteLine("Sum of Matrix");
```

```
for (int i = 0; i < 2; i++)
```

```
{
```

```
    for (int j = 0; j < 2; j++)
```

```
        Console.Write((a1[i, j] + a2[i, j]) + " ");
```

```
    Console.WriteLine();
```

```
}
```

```
// 4. JAGGED ARRAY
```

```
Console.WriteLine("\n4. JAGGED ARRAY");
```

```
int[][] jaggedArray =  
{  
    new int[] { 1, 2, 3 },  
    new int[] { 1, 2, 3, 4 },  
    new int[] { 1, 2 }  
};  
  
foreach (int[] row in jaggedArray)  
{  
    foreach (int n in row)  
        Console.WriteLine(n);  
}  
  
Console.ReadKey();  
}  
}
```

Output:

1. SINGLE DIMENSIONAL ARRAY

abc

lmn

xyz

2. SINGLE DIMENSIONAL ARRAY - FOREACH

pop

up

this

3. TWO DIMENSIONAL ARRAY

Enter Matrix A

1

2

3

4

Enter Matrix B

1

2

3

4

Sum of Matrix

2 4

6 8

4. JAGGED ARRAY

1

2

3

1

2

3

4

1

2

8. SYSTEM.ARRAY CLASS

```
using System;
using System.Collections.Generic;
using System.Text;
namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            // 1. CREATE ARRAY
            Console.WriteLine("1. CREATE ARRAY");

            Array a = Array.CreateInstance(typeof(string), 4);

            a.SetValue("ram", 0);
            a.SetValue("rom", 1);
            a.SetValue("cd", 2);
            a.SetValue("harddisk", 3);

            foreach (string x in a)
                Console.WriteLine(x);

            // 2. ARRAY PROPERTIES
            Console.WriteLine("\n2. ARRAY PROPERTIES");

            Console.WriteLine("Fixed Size: " + a.IsFixedSize);
            Console.WriteLine("Length: " + a.Length);
```

```
Console.WriteLine("Rank: " + a.Rank);
```

```
// 3. LOWER AND UPPER BOUND
```

```
Console.WriteLine("\n3. LOWER AND UPPER BOUND");
```

```
Array b = Array.CreateInstance(typeof(int), 2, 3, 4);
```

```
Console.WriteLine("Lower Bound: " + b.GetLowerBound(1));
```

```
Console.WriteLine("Upper Bound: " + b.GetUpperBound(1));
```

```
Console.WriteLine("Rank: " + b.Rank);
```

```
// 4. SORT
```

```
Console.WriteLine("\n4. SORTED ARRAY");
```

```
Array.Sort(a);
```

```
foreach (string x in a)
```

```
    Console.WriteLine(x);
```

```
// 5. BINARY SEARCH
```

```
Console.WriteLine("\n5. BINARY SEARCH");
```

```
int index = Array.BinarySearch(a, "ram");
```

```
if (index >= 0)
```

```
    Console.WriteLine("Item found at position: " + index);
```

else

Console.WriteLine("Item not found");

// 6. REVERSE

Console.WriteLine("\n6. REVERSED ARRAY");

Array.Reverse(a);

foreach (string x in a)

Console.WriteLine(x);

// 7. CLEAR

Console.WriteLine("\n7. CLEAR ARRAY");

Array.Clear(a, 1, 2);

foreach (string x in a)

Console.WriteLine(x);

// 8. COPY

Console.WriteLine("\n8. COPIED ARRAY");

Array target = Array.CreateInstance(typeof(string), 4);

Array.Copy(a, target, 3);

```
foreach (string x in target)
```

```
    Console.WriteLine(x);
```

```
    Console.ReadKey();
```

```
}
```

```
}
```

```
}
```

9. STRUCTURES & ENUMERATIONS

```
using System;
using System.Collections.Generic;
using System.Text;
struct Book
{
    public string title;
    public string author;
    public int id;
    public int price;
    public double percentage;
}
enum Review
{
    TopSelling = 0,
    DecentCopy = 1,
    FailureCopy = 2
}
namespace ConsoleApplication1
{
    class Program
    {
        static void Main(string[] args)
        {
            Book b = new Book();

            Console.Write("Enter Title: ");
            b.title = Console.ReadLine();
        }
    }
}
```

```
Console.Write("Enter Author: ");
```

```
b.author = Console.ReadLine();
```

```
Console.Write("Enter ID: ");
```

```
b.id = Convert.ToInt32(Console.ReadLine());
```

```
Console.Write("Enter Price: ");
```

```
b.price = Convert.ToInt32(Console.ReadLine());
```

```
Console.Write("Enter Percentage: ");
```

```
b.percentage = Convert.ToDouble(Console.ReadLine());
```

```
Console.WriteLine("\nBook Information");
```

```
Console.WriteLine("Title: " + b.title);
```

```
Console.WriteLine("Author: " + b.author);
```

```
Console.WriteLine("ID: " + b.id);
```

```
Console.WriteLine("Price: " + b.price);
```

```
Console.WriteLine("Percentage: " + b.percentage);
```

```
if (b.percentage < 50)
```

```
    Console.WriteLine("Review: " + Review.FailureCopy);
```

```
else if (b.percentage <= 70)
```

```
    Console.WriteLine("Review: " + Review.DecentCopy);
```

```
else
```

```
    Console.WriteLine("Review: " + Review.TopSelling);
```

```
Console.ReadKey();
```

```
}
```

```
}
```

}

Output:

Enter Title: C# Programming

Enter Author: John

Enter ID: 101

Enter Price: 500

Enter Percentage: 80

Output

Book Information

Title: C# Programming

Author: John

ID: 101

Price: 500

Percentage: 80

Review: TopSelling

If percentage is 60

Book Information

Title: C# Programming

Author: John

ID: 101

Price: 500

Percentage: 60

Review: DecentCopy

If percentage is 40

Book Information

Title: C# Programming

Author: John

ID: 101

Price: 500

Percentage: 40

Review: FailureCopy

10. CLASSES AND OBJECTS

```
using System;

class Student
{
    string name;
    int age;
    int mark1, mark2, mark3;
    public Student(string n, int a, int m1, int m2, int m3)
    {
        name = n;
        age = a;
        mark1 = m1;
        mark2 = m2;
        mark3 = m3;
    }
    public void Display()
    {
        Console.WriteLine("Name: " + name);
        Console.WriteLine("Age: " + age);
        Console.WriteLine("Mark1: " + mark1);
        Console.WriteLine("Mark2: " + mark2);
        Console.WriteLine("Mark3: " + mark3);
    }
}

class Program
{
    static void Main()
    {
        Console.WriteLine("Classes and Objects");
    }
}
```

```
Student s = new Student("Rasika", 21, 90, 92, 88);  
s.Display();  
Console.ReadKey();  
}  
}
```

OUTPUT:

Classes and Objects

Name: Rasika

Age: 21

Mark1: 90

Mark2: 92

Mark3: 88

11. DEFAULT CONSTRUCTORS & PARAMETERIZED CONSTRUCTORS

```
using System;

class Student
{
    int id;
    string name;
    // Default Constructor
    public Student()
    {
        id = 100;
        name = "Bala";
    }
    public void Display()
    {
        Console.WriteLine("Default Constructor:");
        Console.WriteLine(id + " " + name);
    }
}

class Worker
{
    int id;
    string name;
    float salary;
    int experience;
    // Parameterized Constructor
    public Worker(int i, string n, float s, int e)
    {
        id = i;
        name = n;
```

```

        salary = s;
        experience = e;
    }
    public void Display()
    {
        Console.WriteLine(id + " " + name + " " + salary + " " + experience);
    }
}

class Program
{
    static void Main()
    {
        Student s = new Student();
        Worker w1 = new Worker(101, "Raja", 250000f, 5);
        Worker w2 = new Worker(102, "Suresh", 350000f, 3);
        Worker w3 = new Worker(103, "Arun", 250000f, 6);
        Console.WriteLine("DEFAULT CONSTRUCTOR");
        s.Display();
        Console.WriteLine("\nPARAMETERIZED CONSTRUCTOR");
        w1.Display();
        w2.Display();
        w3.Display();
        Console.ReadLine();
    }
}

```

12. SINGLE INHERITANCE

```
using System;

class ProductDetails
{
    public string productName;
    public int id;
    public int quantity;
    public int cost;
}

class Product : ProductDetails
{
    public void GetDetails()
    {
        Console.Write("Enter Product Name: ");
        productName = Console.ReadLine();

        Console.Write("Enter ID: ");
        id = Convert.ToInt32(Console.ReadLine());

        Console.Write("Enter Quantity: ");
        quantity = Convert.ToInt32(Console.ReadLine());

        Console.Write("Enter Cost: ");
        cost = Convert.ToInt32(Console.ReadLine());

        Console.WriteLine("\nProduct Details");

        Console.WriteLine("Product Name: " + productName);
        Console.WriteLine("ID: " + id);
        Console.WriteLine("Quantity: " + quantity);
        Console.WriteLine("Cost: " + cost);
    }
}
```

```
class Program
{
    static void Main()
    {
        Product p = new Product();
        p.GetDetails();
        Console.ReadLine();
    }
}
```

Output:

Enter Product Name: Laptop

Enter ID: 101

Enter Quantity: 5

Enter Cost: 50000

Product Details

Product Name: Laptop

ID: 101

Quantity: 5

Cost: 50000

13. MULTI-LEVEL INHERITANCE

```
using System;

class Mode
{
    public void Mode1()
    {
        Console.WriteLine("There are many modes of transport");
    }
}

class Vehicle : Mode
{
    public void Feature()
    {
        Console.WriteLine("Helps in travelling");
    }
}

class Program : Vehicle
{
    public void Noise()
    {
        Console.WriteLine("All vehicles create noise");
    }

    static void Main()
    {
        Console.WriteLine("Multilevel Inheritance");
        Console.WriteLine("-----");
        Program obj = new Program();

        obj.Mode1();
    }
}
```

```
        obj.Feature();  
        obj.Noise();  
        Console.ReadLine();  
    }  
}
```

Output:

Multilevel Inheritance

There are many modes of transport

Helps in travelling

All vehicles create noise

14. HIEARARCHIAL INHERITANCE

```
using System;

class Library
{
    public string title;
    public string author;
}

class Title : Library
{
    public void GetTitle()
    {
        Console.Write("Enter Title: ");
        title = Console.ReadLine();

        Console.WriteLine("Title = " + title);
    }
}

class Author : Library
{
    public void GetAuthor()
    {
        Console.Write("Enter Author: ");
        author = Console.ReadLine();

        Console.WriteLine("Author = " + author);
    }
}

class Program
```

```
{  
    static void Main()  
    {  
        Console.WriteLine("Hierarchical Inheritance");  
        Console.WriteLine("-----");  
        Title t = new Title();  
        Author a = new Author();  
        t.GetTitle();  
        a.GetAuthor();  
        Console.ReadLine();  
    }  
}
```

Output:

Hierarchical Inheritance

Enter Title: C# Programming

Title = C# Programming

Enter Author: John

Author = John

15. INTERFACE

```
using System;

interface ITransactions
{
    void ShowTransactions();
    double GetAmount();
}

class Transaction : ITransactions
{
    string code;
    string date;
    double amount;
    public Transaction(string c, string d, double a)
    {
        code = c;
        date = d;
        amount = a;
    }
    public double GetAmount()
    {
        return amount;
    }
    public void ShowTransactions()
    {
        Console.WriteLine("Transaction: " + code);
        Console.WriteLine("Date: " + date);
        Console.WriteLine("Amount: " + GetAmount());
        Console.WriteLine();
    }
}
```

```
}  
class Program  
{  
    static void Main()  
    {  
        Transaction t1 = new Transaction("001", "30/07/1997", 9000);  
        Transaction t2 = new Transaction("002", "05/07/1997", 8335);  
        t1.ShowTransactions();  
        t2.ShowTransactions();  
        Console.ReadLine();  
    }  
}
```

Output:

Transaction: 001

Date: 30/07/1997

Amount: 9000

Transaction: 002

Date: 05/07/1997

Amount: 8335

16. EXCEPTION HANDLING

```
using System;

class Program
{
    static void Main()
    {
        try
        {
            Console.Write("Enter age: ");

            int age = Convert.ToInt32(Console.ReadLine());

            if (age < 18)
                throw new Exception("Age must be 18 or above");

            Console.WriteLine("Eligible");
        }
        catch (Exception e)
        {
            Console.WriteLine("Exception: " + e.Message);
        }
        finally
        {
            Console.WriteLine("Finally block executed");
        }

        Console.ReadLine();
    }
}
```

Output

If age is 20:

Enter age: 20

Eligible

Finally block executed

If age is 15:

Enter age: 15

Exception: Age must be 18 or above

Finally block executed

17. COUNTING ALPHABETS USING FOREACH STATEMENT

```
using System;
namespace foreach
{
    class Program
    {
        static void Main(string[] args)
        {
            string s;
            int a = 0, t = 0, space = 0;
            Console.WriteLine("Enter a string");
            Console.WriteLine("-----");
            s = Console.ReadLine();
            Console.WriteLine("-----");
            Console.WriteLine("Result is:");
            foreach (char ch in s)
            {
                if (ch == 'a')
                    a++;
                if (ch == 't')
                    t++;
                if (ch == ' ')
                    space++;
            }
            Console.WriteLine("No of 'A' = " + a);
            Console.WriteLine("No of 'T' = " + t);
            Console.WriteLine("No of 'Space' = " + space);
            Console.ReadLine();
        }
    }
}
```

```
}  
}
```

Output:

Enter a string

data table

Result is:

No of 'A' = 3

No of 'T' = 2

No of 'Space' = 1